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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	KAI
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG Track 8, 9, 12
Project / Solution Name	One Stop Smart Procurement System
Name (Leader)	Woon Kai Xiang
Email & Phone (Leader)	woonkx911@gmail.com 016-7230900
University Affiliation(s)	Sunway University
Number of Team Members	1

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Our team is tackling the difficulty SMEs face when sourcing urgent, specialist spare parts. This is especially common in car repair shops, where workers may need parts for many vehicles makes and models but have limited space to keep every item in stock. When a part is unavailable, employees and owners may spend hours checking inventory, searching multiple suppliers, requesting quotes and comparing prices, often with incomplete information about fit, availability or delivery time.

This affects shop owners, mechanics and procurement staff, as well as customers who may wait longer for repairs. **Economically**, delays can mean lost working hours, postponed jobs, higher urgent-purchase costs and money tied up in unsuitable or excess stock. **Socially**, repeated manual searching adds pressure to small teams and can affect the service they provide. **Environmentally**, buying too much slow-moving or incorrect stock can contribute to unused parts and waste.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

One Stop Smart Procurement System is a web app that guides an SME through a purchase request: an employee enters the part and when it is needed; the system checks existing inventory and supplier records; staff can search for other suppliers and compare available information; a manager approves the purchase; and the team records its status in an activity log.

AI is used as a decision-support assistant. It can summarize recorded quotations, explain the reasons for a recommendation and help staff compare options. Supplier prices and stock are not treated as confirmed just

because they appear in a search result: an employee must confirm the quote, and a manager makes the approval decision.

From the SDC Phase 1 workshops, I intend to apply skills in using Replit to build and iterate on an app, and writing clear AI prompts that describe the problem, users, requirements and constraints. I will also use an iterative process—reviewing the generated app, testing the workflows and refining the prompts and implementation based on what works.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

The current prototype uses:

- **Frontend:** React, TypeScript and Vite, with Tailwind CSS and reusable UI components.
- **Backend:** Node.js with Express and TypeScript, exposing a REST API for procurement actions.
- **Database:** PostgreSQL, accessed with Drizzle ORM. Request, inventory, supplier, quotation, approval and audit information is stored in the workspace.
- **AI:** Replit AI Integrations' OpenAI-compatible API, using **GPT-5.4 mini** to explain comparisons based on recorded quotations. Supplier scoring is rules-based; the AI does not invent prices or make the approval decision.
- **API Usage:** SerpApi's Google Search and Google Maps results help find potential supplier leads. Staff still need to confirm a quote.
- **Cloud platform:** Replit is used to develop and run the web app and API. Employee and manager roles are simulated in the demo; purchases are tracked internally, not sent to suppliers.

SDC sponsor-provided resources:

- **Replit**

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

My design thinking began with firsthand experience helping at my parents' SME automotive workshop. When a customer needed a rare-model spare part that wasn't in the inventory or supplier catalog, I watched my mother spend hours finding suppliers, confirming prices, placing orders and recording details in a separate Excel folder. This showed me the problem wasn't just finding a part: information was fragmented, and manual handoffs made sourcing slow and difficult to track.

I defined the need as a clearer process connecting a worker's request to available stock, supplier options, quote confirmation and manager approval. I then prototyped Smart Procurement to bring those steps together, while keeping people responsible for confirming quotes and approving purchases.

Phase 2, I would test it with workshop staff, compare sourcing time before and after, and gather feedback on usability. Before real deployment, I would also verify secure staff and manager access, data accuracy and any required system integrations.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: SDG Target 8.3 — support productive activities, entrepreneurship, innovation and the growth of micro-, small- and medium-sized enterprises. Secondary potential: SDG Target 12.5 — reduce waste through prevention, reduction, recycling and reuse. The inventory check may help prevent unnecessary purchases, but that impact has not yet been measured.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	1.5 active staff hours per uncommon-part request to search for suppliers and confirm a quote.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Median active staff **hours** per uncommon-part request, from recording the need to recording a supplier-confirmed quote.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation*	Reduce the median active time by 30% for recording the need to recording a supplier-confirmed quote.

Metric Category	Your Response
e.g., "20% improvement in computational resource use."	

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ **No**

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React	Frontend framework	MIT
TypeScript	Programming language/tooling	Apache-2.0
Vite	Frontend build tool	MIT
Express	Backend API framework	MIT
Drizzle ORM	Database library	Apache-2.0
PostgreSQL and `pg` driver	Database and code library	PostgreSQL License / MIT
Tailwind CSS	Styling framework	MIT
Radix UI, TanStack Query and Zod	UI components, data fetching and validation libraries	MIT
Clerk SDK	Authentication library	MIT
OpenAI SDK	Code library for AI API access	Apache-2.0

GPT-5.4 mini via Replit AI Integrations	Pre-trained AI model/service	Proprietary hosted service; governed by provider terms
SerpApi with Google Search and Maps results	Supplier discovery API and live external data	Provider terms; not an open-source dataset
Sample supplier, inventory and quotation records	Demo dataset	Team-created; no third-party dataset license

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] ***KAI***

[Date] 24/10/2026